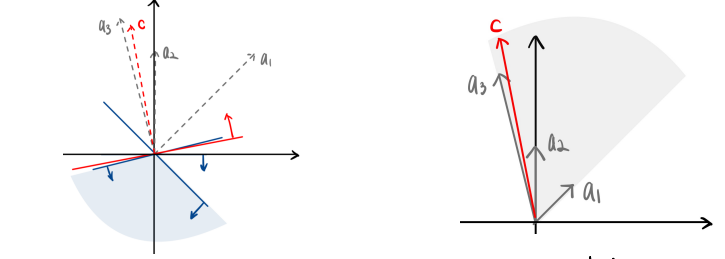


Farkas' Lemma 둘 중 하나만 해가 존재한다.

Sys 1: $AX \geq 0$ and $CX < 0$
 Sys 2: $WA = C$ and $w \geq 0$
 (System 1 sol 존재 X) (System 2 sol 존재)

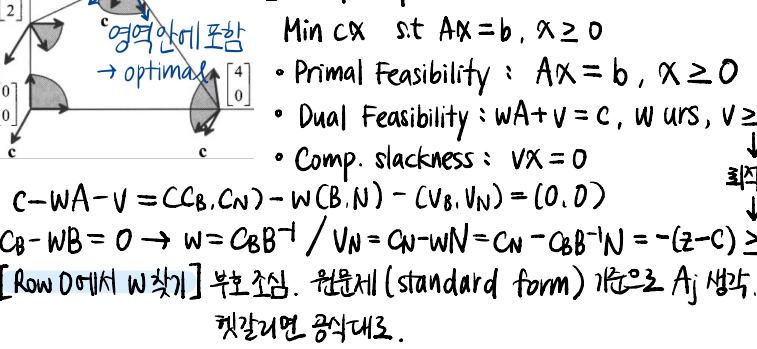


KKT Optimality Conditions : 최적상대의 필요충분조건

[for Inequality Constrained Problems] canonical form에서 구성

Min CX
 s.t $AX \geq b$
 $x \geq 0$

- Primal Feasibility : $AX \geq b, x \geq 0$
- Dual Feasibility : $WA + v = c, w \geq 0, v \geq 0$
- Comp. slackness : $w(AX - b) = 0, vx = 0$



Min CX s.t $AX = b, x \geq 0$
 • Primal Feasibility : $AX = b, x \geq 0$
 • Dual Feasibility : $WA + v = c, w$ urs, $v \geq 0$
 • Comp. slackness : $vx = 0$

$C - WA - v = CC_B, CN - w(CB, N) - (v_B, v_N) = (0, 0)$
 $CB - WB = 0 \rightarrow w = CB^{-1} / VN = CN - WN = CN - CB^{-1}N = -(z - C) \geq 0$
 [Row 0에서 w 찾기] 부호조심. 원문제 (standard form) 기준으로 A_j 생각.
 켄달리면 공식대로.

Duality : One dual var for each primal const. (가르치드)

CP) Min $CX \leftrightarrow$ (CD) Max wb s.t $AX \geq b, x \geq 0$
 CP) Min $CX \leftrightarrow$ (CD) Max wb s.t $WA \leq C, w \geq 0$

	Minimization Problem		Maximization Problem	
Variables	≥ 0	$\iff$	$\leq$	Constraints
	≤ 0	$\iff$	$\geq$	
	Unrestricted	$\iff$	$=$	
Constraints	$\geq$	$\iff$	≤ 0	Variables
	$\leq$	$\iff$	≥ 0	
	$=$	$\iff$	Unrestricted	

Weak Duality The obj. func. value for any feasible sol to the min. prob. is always gq. to the obj. func. value for any feasible sol to the max. prob. (MIN $\geq$ MAX $\rightarrow$ 서로의 LB/UB)
 $\rightarrow CX_0 = w_0b$ 이면 각자의 opt. solution 이고 무조건 다 optimal. 목적함수값 동일.

P	Optimal	$\iff$	D	Optimal
P(D)	Unbounded	$\Rightarrow$	D(P)	Infeasible
P(D)	Infeasible	$\Rightarrow$	D(P)	Unbounded or Infeasible
P(D)	Infeasible	$\iff$	D(P)	Unbounded in Homogeneous Form

(HD) max wb s.t. $WA \leq 0, w \geq 0$

Complementary Slackness & Supervisor's Principle
 $\rightarrow (C_j - w^*a_j)x_j^* = 0 \forall j$ and $w_j^*(a_jx^* - b_i) = 0 \forall i$
 둘 중 하나는 0이어야 함

opt. sol. 을 본인 제약식에 대입. non-binding 하면 다른 쪽 var = 0.
 binding 하면 다른 쪽 var 값 존재.

Economic Interpretation of the Dual.

Assume a company should produce $b_1, b_2, \dots, b_m$ units of m products.

- The company can produce these outputs by engaging in any of n activities (production methods for example) at varying levels. CP) Min $\sum c_j x_j$ s.t $\sum a_{ij} x_j \geq b_i$
- a_{ij} is the units of product i produced by one unit of activity j .
- If c_j is the unit cost of activity j then the company can minimize its production cost by solving the following LP in which x_j is the level of activity j .

The objective is maximization of the seller's profit. The constraint make sure that the price of paid to the seller for the products that can be produced by one unit of activity j is less than the cost of actually performing one unit of activity j to produce the products. This will keep the company willing to buy from the seller.

If z^* is the optimal objective function value and B_i^{-1} is the i th column of B^{-1} ,

$$\frac{\sigma z^*}{\sigma b_i} = c_B B_i^{-1} = w_i^*.$$

Thus, w_i^* is the rate of change of the optimal objective value with a unit increase in the i th RHS value, given that the current nonbasic variables are held at zero.

Economically, we may think of w^* as a vector of shadow prices for the RHS vector, which is the incremental cost of producing one more unit of the i th product or the fair price we would pay to have an extra unit of the i th product.

Assume that problem P is a product-mix problem in which n products are being manufactured using some m types of resources. CP) Max $\sum c_j x_j$ s.t $\sum a_{ij} x_j \leq b_i$

Observe that if the manufacturer considers renting out the m resources at unit prices $w_1, \dots, w_m$ instead of manufacturing the mix of products. CP) Min $\sum b_i w_i$ s.t $\sum a_{ij} w_i \geq c_j$

Then every unit of product j not manufactured would result in a loss of profit of c_j units but would earn a rent of $\sum_{i=1}^m a_{ij} w_i$ units by virtue of the resources this releases.

Primal Opt. 자체인 상태에서 Primal Feas 찾기

The Dual Simplex Method - 결론 Opt 2 dual unb. (= primal inf.)

- 1) Dual Feas. (= Primal Opt) 성립하는지 Row 0에서 확인 $z_j - c_j \leq 0$ $\forall j$ min
- 2) Basis 구성 잘 되었는지 확인하기 $\rightarrow$ IR 조건 확인
- 3) leaving var 선택: RHS < 0 중 가장 작은 값 선택.
- 4) pivot (entering var 선택): $y_{rj} < 0$ 중에서, $\min \{ \frac{z_j - c_j}{y_{rj}} \}$
- 5) w 찾기 $\rightarrow$ initial Basis가 되었다면, w는 e_j 에 대응되게 존재. a_j 부호조심.

Initial Dual Feasible Solution : Dual 실패스쓰고 싶는데 row 0 > 0 존재

Suppose that the first m columns constitute the initial basis. If $z_j - c_j > 0$ for some j , add the constraint $\sum_{j=m+1}^n x_j \leq M$, where $M > 0$ is large, to the problem. Then, we have the following tableau, where x_{n+1} is the slack variable of the additional constraint.

	Z	x_1	x_2	$\dots$	x_m	x_{m+1}	$\dots$	x_n	x_{n+1}	RHS
Z	1	0	0	$\dots$	0	$z_{m+1} - c_{m+1}$	$\dots$	$z_n - c_n$	0	$c_B b$
x_{n+1}	0	0	0	$\dots$	0	1	$\dots$	1	1	M
x_1	0	1	0	$\dots$	0	$y_{1,m+1}$	$\dots$	y_{1n}	0	b_1
x_2	0	0	1	$\dots$	0	$y_{2,m+1}$	$\dots$	y_{2n}	0	b_2

$\rightarrow$ pivot 해서 row 0가 다 < 0 이 되도록 pivot 진행해고 시작.

Sensitivity Analysis

[Change in the cost vector] $C_k \rightarrow C'_k$ (dual feas 위반가능)

- ① x_k is nonbasic : $z_k - C'_k = (z_k - C_k) - (C_k - C'_k) < 0$: optimal 유지
- ② x_k is basic : Row 0 다 다시 계산 ≥ 0 : primal simplex

(j=k) $z_k - C'_k = 0$ 고정 (j≠k) $z_j - c_j = (z_j - c_j) + (C'_k - C_k) y_{kj} > 0$: primal simp.

[Change in the RHS vector] $b \rightarrow b'$: $B^{-1}b' = B^{-1}b + \sum y_j (b'_j - b_j)$ $B^{-1}b' \geq 0$: optimal 유지. / $B^{-1}b' \leq 0$: dual simplex 진행.

[Addition of a new activity] $x_{n+1}, C_{n+1}, a_{n+1}$ 소개 $\rightarrow x_{n+1}$ worthwhile? $z_{n+1} - C_{n+1} = w a_{n+1} - C_{n+1} < 0$: 현재 해 opt. $x_{n+1} = 0$ > 0 : primal simplex. $y_{n+1} = B^{-1}a_{n+1}$ 계산

[Addition of a new constraint] $a^{m+1}x \leq b^{m+1}, a^{m+1}x + x_{n+1} = b^{m+1}$ x_{n+1} 이 slack으로 BV가 되도록 ER0 진행해서 정리 $\rightarrow$ (dual) simplex 진행.

[Change in the constraint matrix A] $a_j \rightarrow a'_j$

- 1) for nonbasic col. : $z'_j - c_j = c_B B^{-1}a'_j - c_j \leq 0$: opt. / ≥ 0 : simplex
- 2) for basic col. : ① assume x'_j is added with a'_j and c_j
 $\rightarrow y'_j = B^{-1}a'_j$ and $z'_j - c_j = c_B B^{-1}a'_j - c_j$ (B. 현재꺼)
 ② orig. x_j 삭제 $\begin{cases} y'_{jj} \neq 0 \rightarrow$ 바로 삭제 가능. $\rightarrow$ ER0 basis 만들기
 $y'_{jj} = 0 \rightarrow$ orig x_j 에 art. var (-M) row 0에 추가

Integer Programming : $S \subset \text{conv}(S) \subset \text{LP relaxation}$
 $S = \{x \in \mathbb{Z}^n : Ax \leq b\}$ = feasible region $\rightarrow \text{conv}(S)$: convex comb. of S
 IP $\rightarrow$ LP relaxation $\rightarrow$ opt. sol $\rightarrow$ (int?) yes $\rightarrow$ opt.
 (if no, then add constraints and repeat)

Preprocessing
[Tightening Bounds]

$y_1 + y_2$	$= 1.46$ ①	$y_1 \leq 1.46x_1, \quad y_2 \leq 1.46x_2$ ①
$y_3 + y_4$	$= 0.72$ ②	$y_3 \leq 0.72x_3, \quad y_4 \leq 0.72x_4$ ②
$-y_2 - y_3 + y_5$	$= 0$ ③	$y_5 \leq (1.46 + 0.72)x_5, \quad y_6 = 0.32$ ④

[Adding logical inequalities] **[Fixing variables]**
[Removing redundant constraints]

Replace x_i by $1 - x_i'$ for $a_i < 0$

$$\begin{aligned} -3x_2 - 2x_3 &\leq -2 \rightarrow -3(1-x_2') - 2x_3 \leq -2 \rightarrow -3 + 3x_2' - 2x_3 \leq -2 \rightarrow 3x_2' - 2x_3 \leq 1 \\ -4x_1 - 3x_2 - 3x_3 &\leq -6 \rightarrow -4x_1 - 3(1-x_2') - 3x_3 \leq -6 \rightarrow -4x_1 - 3 + 3x_2' - 3x_3 \leq -6 \rightarrow -4x_1 + 3x_2' - 3x_3 \leq -3 \\ 2x_1 - 2x_2 + 6x_3 &\leq 5 \rightarrow 2x_1 - 2(1-x_2') + 6x_3 \leq 5 \rightarrow 2x_1 - 2 + 2x_2' + 6x_3 \leq 5 \rightarrow 2x_1 + 2x_2' + 6x_3 \leq 7 \end{aligned}$$

$x \in \mathbb{B}^3$ ④ bi decoding 필요

Decomposition Algorithm

(Primal) Min $Cx \rightarrow$ (Master) Min $\sum_{j=1}^n (cx_j)\lambda_j + \sum_{j=1}^m (cd_j)\mu_j$
 s.t. $Ax = b$ $x \in X$ λ 문제 $\rightarrow$ s.t. $\sum_{j=1}^n (Ax_j)\lambda_j + \sum_{j=1}^m (Ad_j)\mu_j + s = b \rightarrow w$
 $\sum_{j=1}^n \lambda_j = 1 \rightarrow d$
 $\lambda_j, \mu_j \geq 0$
 LB: $\hat{C}_B \bar{b} - (z_k - \hat{C}_k) \rightarrow 0$ $\rightarrow$ UB
 $x^* = \sum_{j \in \text{basic}} \lambda_j x_j + \mu_j d_j = \lambda_k x_k + \mu_k d_k$
 optimal Master problem 해대로 sub. problem 해 가져와주기
 (Sub) Max $(wA - C)x + d$ s.t. $x \in X$
 λ_k 문제 $\rightarrow$ $\rightarrow z_k - \hat{C}_k$

절차 / 유의사항

- 1) Primal에서 $Ax \leq b$ 란 X 구별 [$Ax \leq b$ 에서 slack 포함]
 X bounded 인지 unbounded 인지 구별
- 2) Master 구성 (initiate) [basis에 S 란 λ_1 포함하여 구성 $\rightarrow B = B^{-1} = I$
 initial BFS x_1 구성, (w, d) 계산 = $Cx_1 B^{-1}$
 $\bar{b}$ 계산 = $B^{-1} \begin{bmatrix} b \\ 1 \end{bmatrix}$, $z = \hat{C}_B \bar{b}$ (=UB)
- 3) Iteration

$w \leq 0$ 면 slack (바닥) enter 불가. $w > 0$ 이면 revised update 바로.

[Sub] Max $(wA - C)x + d \rightarrow A, C$: primal m , w, d : Master m
 s.t. $x \in X$ $\rightarrow$ 원점!
 $\rightarrow$ simplex / graphically 풀기 : x_2 (d2) 구성 / 기각. $z_2 - \hat{C}_2$ 구하기 $< d : (wA - C)d$
 Row 0 가 음수 인게 entering LB = $\hat{C}_B \bar{b} - (z_2 - \hat{C}_2)$ 구하기.
 $\rightarrow$ 최적 아니면, λ_2 (μ_2) 구성. $\lambda_2 = \begin{bmatrix} z_2 - \hat{C}_2 \\ y_2 \text{ 계산} \rightarrow B^{-1}(Ax_2) \end{bmatrix}$ or $\mu_2 = \begin{bmatrix} z_2 - \hat{C}_2 \\ y_2 \text{ 계산} \rightarrow B^{-1}(Ad_2) \end{bmatrix}$

[Master] revised simplex tableau 업데이트 ($\sum \lambda = 1$ 인지 확인)
 UB 업데이트. (w, d) 업데이트
 $w \leq 0$ 면 slack (바닥) enter 불가. $w > 0$ 이면 revised update 바로

[Sub] Max $(wA - C)x + d \rightarrow A, C$: primal m , w, d : Master m
 s.t. $x \in X$
 $\rightarrow x_3$ 구성. $z_3 - \hat{C}_3$ 계산 $\rightarrow$ if $z_3 - \hat{C}_3 \leq 0$ 이면 최적
 전 Master 테이블 = optimal.
 Primal $x^* = \sum \lambda_k x_k$

Inequality constraints (continued)

- Suppose that there is no obvious $x \in X$ with $Ax \leq b$. In this case, after converting the master problem to equality form by adding appropriate slack variables, the constraints are manipulated so that the RHS values are nonnegative.
- Then artificial variables are added, as needed, to create an identity matrix. This identity matrix constitutes the starting basis. The two-phase or big-M methods can be used to drive the artificial variables out of the basis.

Equality constraints

- $m + 1$ artificial variables can be introduced to form the initial basis, and the two-phase or the big-M method can be applied.

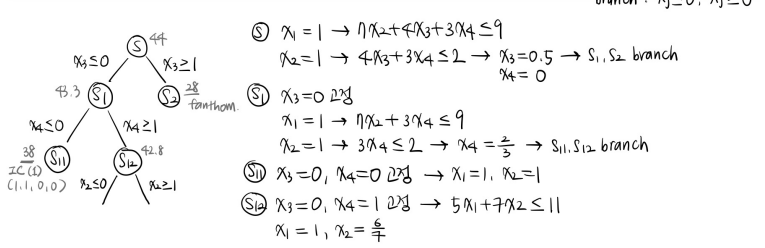
Branch and Bound (Max 기준)

- parent UB $\geq$ child UB 기준 / - IC $\geq$ UB면 prune
- integer feasible 해야 IC 평가가능.

0-1 knapsack Problem max $\sum_{j=1}^n c_j x_j$ s.t. $\sum_{j=1}^n a_j x_j \leq b, x_j \in \{0, 1\}^n$
 $\rightarrow x_j$ 한단위에 $\frac{c_j}{a_j}$ worth $\rightarrow$ 제일큰 worth가인 $x_j = 1$ 고정 $\rightarrow$ 제약씩 채워가면서

Example 8.6 A Knapsack Problem

x_j 한단위에	$\frac{16}{5}$	$\frac{22}{7}$	$\frac{12}{4}$	$\frac{8}{3}$
Max	$16x_1 + 22x_2 + 12x_3 + 8x_4$			
s.t.	$5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14$			
	$x_1, x_2, x_3, x_4 \in \mathbb{B} \rightarrow 0 \leq x_1, x_2, x_3, x_4 \leq 1$			
	branch : $x_j \leq 0, x_j \geq 0$			



	BASIC INVERSE	RHS
z	$(w, d) = \hat{C}_B B^{-1}$	$\hat{C}_B \bar{b} = C_B x_j B^{-1} b$
S	B^{-1}	$\bar{b} = B^{-1} \begin{bmatrix} b \\ 1 \end{bmatrix}$
λ		

$\bar{b}_r = \min_{1 \leq i \leq m} \left\{ \frac{\bar{b}_i}{y_{ik}} : y_{ik} > 0 \right\}$
 $\rightarrow$ leaving var 결정
 $\rightarrow$ pivot (ERO)

(Simplex)

Bounded { if $z_k - \hat{C}_k = 0$: 전 Master BFS (λ_0, λ_1)가 opt.
 if $z_k - \hat{C}_k > 0$: λ_k enters. $y_k = B^{-1} \begin{pmatrix} Ax_k \\ 1 \end{pmatrix} \rightarrow \begin{bmatrix} z_k - \hat{C}_k \\ y_k \end{bmatrix}$
 $\rightarrow x \notin X$ 인 slack에 대해서 $z_{s_i} - C_{s_i} = w_i > 0$ 이면 enter 가능

Unbounded { if sub = bounded $\begin{bmatrix} z_k - \hat{C}_k = (w, d) \begin{pmatrix} Ax_k \\ 1 \end{pmatrix} - Cx_k \leq 0 \\ = (w, d) \begin{pmatrix} Ad_k \\ 0 \end{pmatrix} - cd_k \leq 0 \end{bmatrix}$ optimal
 $z_k - \hat{C}_k > 0$: (x_k 이) λ_k enter $\begin{bmatrix} z_k - \hat{C}_k \\ B^{-1}(Ax_k) \end{bmatrix}$
 if sub = unbounded : (d_k 이) μ_k enter $\begin{bmatrix} (wA - C)d_k \\ B^{-1}(Ad_k) \end{bmatrix}$

Steps of the Simplex Method (Minimization Problem).

1. The basic feasible solution is given by $x_B = B^{-1}b = \bar{b}$ and $x_N = 0$. The objective $z = c_B B^{-1}b = c_B \bar{b}$.
2. Calculate the simplex multipliers $w = c_B B^{-1}$. For each nonbasic variable, calculate $z_j - c_j = c_B B^{-1}a_j - c_j = wa_j - c_j$. Let $z_k - c_k = \max_{j \in N} z_j - c_j$. If $z_k - c_k \leq 0$, then stop; the current solution is optimal. Otherwise, go to step 3.
3. Calculate $y_k = B^{-1}a_k$. If $y_k \leq 0$, then stop; the optimal objective value is unbounded. Otherwise, determine the index of the variable x_{B_r} leaving the basis as follows:

$\frac{\bar{b}_r}{y_{rk}} = \min_{1 \leq i \leq m} \left\{ \frac{\bar{b}_i}{y_{ik}} : y_{ik} > 0 \right\}$

To proceed with the steps of the Simplex method, we form the following tableau,

	z	x_B	x_N	RHS
z	1	0	$c_B B^{-1}N - c_N$	$c_B B^{-1}b$
x_B	0	I	$B^{-1}N$	$B^{-1}b$

Row 0
Rows 1 through m